

CURRICULUM VITAE

Dongwook Kwon

AI Engineer · Intelligent Agent Technology

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Statement of Areas of Expertise

Dongwook Kwon is an AI engineer and graduate researcher whose work centers on **agentic AI systems, multi-agent large language model (LLM) evaluation**, and **time-series anomaly detection** for safety-critical domains such as implantable medical devices, power-generation infrastructure, and cyber-physical systems. His research has been published in leading international venues including the Association for Computational Linguistics (ACL) and the Empirical Methods in Natural Language Processing (EMNLP) conferences, both ranked among the top peer-reviewed venues in natural language processing globally [TODO: confirm/cite CORE A-star / CSrankings tier]. He is the holder of two granted Republic of Korea patents addressing critical problems in implantable medical device safety and network-anomaly detection, and the recipient of national-level awards from the Korean Ministry of Oceans and Fisheries and the United States-based Qualcomm Institute at UC San Diego.

EDUCATION

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| 2024–2026 | Master of Science in Computer Engineering
Kwangwoon University, Seoul, Republic of Korea
Grade Point Average: 4.33 / 4.50 (98.1%)
Thesis advisor: Prof. Cheolsoo Park. Affiliated with the Bio-Computing and Machine Learning Laboratory. |
| Jan–Jul 2025 | Graduate Research Program in Mechanical and Industrial Engineering
University of Toronto, Toronto, Ontario, Canada
Grade Point Average: 3.68 / 4.00
Visiting graduate researcher under a joint program with LG Electronics Canada. |
| 2019–2024 | Bachelor of Science in Electronics and Communications Engineering
Kwangwoon University, Seoul, Republic of Korea
Grade Point Average: 3.41 / 4.50 (88.1%). |

PROFESSIONAL EXPERIENCE

- Dec 2025 – Present** **AI Engineer**, Intelligent Agent Technology Team
Suresoft Technologies, Seongnam, Gyeonggi, Republic of Korea
Leading development of an AI validation system built on collaborative AI-agent architectures. Designing and implementing infrastructure to verify and evaluate machine-learning models through autonomous and cooperative AI agents. Building agentic AI systems that enable systematic, scalable, and reliable AI verification – addressing a pressing industry need as enterprise deployments of LLMs scale.
- Aug 2021 – Dec 2025** **Research Assistant**
Bio-Computing and Machine Learning Laboratory, Kwangwoon University, Seoul, Republic of Korea
Conducted research on deep-learning algorithms with a focus on anomaly detection in time-series data, including electrocardiogram (ECG) signals and cyber-physical-system telemetry. Contributed substantively to the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening Internet-of-Things medical devices – a system designed to prevent catastrophic outcomes from cyber compromise of cardiac implantable devices.
- Jan – Jul 2025** **Visiting Researcher (Research Project)**
LG Electronics Canada, Toronto, Ontario, Canada
Joint program with the University of Toronto. Contributed to the development of an LLM-driven assessment framework that systematically evaluates the performance of agentic AI systems – work that fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.
- Jul – Aug 2022** **AI Development Project Participant**, Summer Program
Qualcomm Institute, University of California San Diego, San Diego, California, United States
Selected for a competitive international summer program. Contributed to the Personality-Based Drug Addiction Prediction System project, applying machine-learning techniques to behavioral and personality data. Recognized with the program's Achievement Award.

HONORS AND AWARDS

- Dec 2025** **Silver Award**, 23rd Embedded Software Competition, Korean Institute of Information Scientists and Engineers (KIISE), Republic of Korea. A nationally recognized competition in embedded software engineering. [TODO: number of participants / award rate]
- Nov 2025** **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers (IEIE), Republic of Korea.

- Nov 2024** **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC), Republic of Korea.
- Nov 2023** **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering (KOSOMBE), Republic of Korea.
- Dec 2022** **Bronze Award**, Hanium ICT Mentoring Competition, Federation of Korean Information Industries (FKII), Republic of Korea.
- Sep 2022** **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX), Kwangwoon University, Republic of Korea.
- Aug 2022** **Achievement Award**, Qualcomm Institute AI Development Project, University of California San Diego, United States. Awarded for outstanding contributions in a competitive international summer program.
- Dec 2021** **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project, Minister of Oceans and Fisheries, Republic of Korea. This award is conferred by a cabinet minister of the Republic of Korea and represents a nationally significant recognition of contribution to maritime logistics technology.
- Oct 2021** **Grand Prize**, MY (Multi-Y) Capstone Design Competition, Kwangwoon University, Republic of Korea. Top award among all participating senior engineering capstone teams.

PEER-REVIEWED PUBLICATIONS

Journal Articles

1. Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park. “Designing a Remote Photoplethysmography-Based Heart Rate Estimation Algorithm during a Treadmill Exercise.” *Electronics*, vol. 14, no. 5, Article 890, 2025.
Publisher: MDPI. [TODO: journal impact factor, citation count if available]
2. **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park. “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments.” Under peer review at *Data Mining and Knowledge Discovery* (Springer Nature), 2025.
[TODO: journal impact factor (DMKD is a leading journal in data mining)]

Conference Papers

1. R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma. “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades.” In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL) – Industry Track (Poster)*, 2026.
[†]Equal contribution.
ACL is the flagship conference in computational linguistics and natural language processing.
[TODO: acceptance rate of industry track]
2. J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma. “GEMMAS: Graph-Based Evaluation Metrics for Multi-Agent Systems.” In *Proceedings of EMNLP 2025 – Industry Track (Oral Presentation)*, 2025.
[†]Equal contribution.

EMNLP is a top-tier conference in empirical methods for NLP. The work was selected for oral presentation. [TODO: oral acceptance rate]

3. **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park. “Hybrid Neural Network Model for Anomaly Detection in Implantable Devices Using Graph Attention Networks and Transformers.” In *Proceedings of the 10th International Biomedical Engineering Conference (IBEC)*, 2024. *Best Poster Award – Silver Winner*.
4. **D. Kwon**, Y. Kang, and C. Park. “Enhancing Medical Device Security with GNN-GRU Anomaly Detection Model.” In *Proceedings of the 62nd KOSOMBE Autumn Conference*, pp. 204–205, 2023. *Outstanding Paper Award*.

PATENTS

1. “Method for Implantable Medical Device to Detect Anomaly in Real Time,” Republic of Korea Patent No. **KR 10-2961984**, issued April 30, 2026.
Addresses real-time anomaly detection for cardiac implantable devices – a safety-critical capability for the rapidly growing population of pacemaker and ICD recipients.
2. “GPT API-Based Network Anomaly Detection,” Republic of Korea Patent No. **KR 10-2848814**, issued August 18, 2025.
Applies large-language-model APIs to the problem of network-traffic anomaly detection – a novel methodology bridging generative AI with classical cybersecurity.

FUNDED RESEARCH PROJECTS

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| 2025 – Present | Agentic AI Evaluation Framework – LG Electronics Canada
Developed an evaluation framework for agentic AI systems leveraging LLMs to assess performance and reliability. |
| 2022 – Present | AI-Based Intrusion Detection and Attack-Packet Crafting Technologies for APR1400 Nuclear Reactors – Sejong University, with national infrastructure stakeholders
Developed an AI-based defense system protecting nuclear power plants from cyber attacks. The APR1400 is a generation-III+ pressurized water reactor design deployed in the Republic of Korea and exported internationally. |
| Apr – Nov 2022 | Quadruped Robot Using Vision SLAM with a Depth Camera – Ministry of Science and ICT, Republic of Korea
Developed an autonomous quadruped robot using a navigation algorithm with simultaneous localization and mapping (SLAM) and a depth camera. |
| Jul – Aug 2022 | Personality-Based Drug Addiction Prediction System – Qualcomm Institute, University of California San Diego
Developed a personality-based drug-addiction prediction system using machine-learning models. Selected international participant. |

- 2021 – 2022 Active Kill-Switch and Biomarker-Based Defense System for Life-Threatening IoT Medical Devices** – Kookmin University, industry-academia collaboration
Developed a security system for cardiac implant devices using anomaly- detection algorithms.
- Apr – Nov 2021 Smart Logistics System Using Computer Vision** – Ministry of Oceans and Fisheries, Republic of Korea
Developed a logistics-detection system incorporating location tracking and stack-layer identification for shipping containers. Resulted in the Ministerial Gold Award.
- May – Oct 2021 Smart Kitchen IoT System Utilizing Computer Vision and Deep Learning** – Kwangwoon University
Developed a cloud IoT platform with a grocery-classification system and an OCR receipt program.

LEADERSHIP AND SERVICE

- **President**, Kwangwoon International Student Association – led intercultural programs and student-engagement initiatives spanning multiple cohorts of international students.
- **Director of Filming and Editing**, Kwangwoon Broadcasting Center (KWBC) – oversaw media production and content delivery for the campus broadcaster.
- **Class Representative**, Department of Electronics and Communications Engineering – represented student interests in academic affairs.
- **Executive Member**, Amateur Radio Club – organized radio workshops and technical exhibitions.

LANGUAGES

- **Korean** – Native proficiency.
- **English** – Professional working proficiency (completed graduate research programs in Toronto, Canada and San Diego, California; published in English-language peer-reviewed venues).

SELECTED SKILLS AND TECHNICAL COMPETENCIES

- **Machine learning and deep learning**: PyTorch, TensorFlow; transformers, graph neural networks, sequence models (GRU, LSTM); anomaly detection; multi-agent systems; retrieval-augmented generation.
- **Software engineering**: Python (primary), C/C++, MATLAB; Linux, Docker, Git; CUDA programming.
- **Domain experience**: Biomedical signals (ECG, photoplethysmography), cyber-physical systems, industrial control / SCADA, embedded systems.

ADDITIONAL INFORMATION

[TODO: To strengthen the NIW petition, the following items are worth adding once available:]

- **Citation counts and h-index:** provide Google Scholar profile metrics.
 - **Invited talks and presentations:** list any keynote, invited, or conference-talk engagements.
 - **Editorial and review service:** any peer-review for journals or conferences.
 - **Professional society memberships:** e.g., ACL, IEEE, ACM.
 - **Media coverage:** any press or media coverage of work, particularly the awards (Ministerial Gold, IBEC, ACL/EMNLP).
 - **Letters of recommendation:** collected separately; this CV should be consistent with statements made by recommenders.
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End of Curriculum Vitae · Dongwook Kwon